

Thinking Like a Computational Materials Scientist

Testing Scientific Hypotheses with Machine Learning



Brookhaven
National Laboratory



U.S. DEPARTMENT
of ENERGY

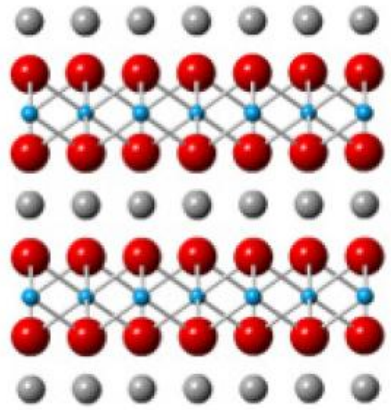
Sara E. Mason
Group Leader, Theory & Computation
Center for Functional Nanomaterials

WDTs Big Data Science and Applications Summer School

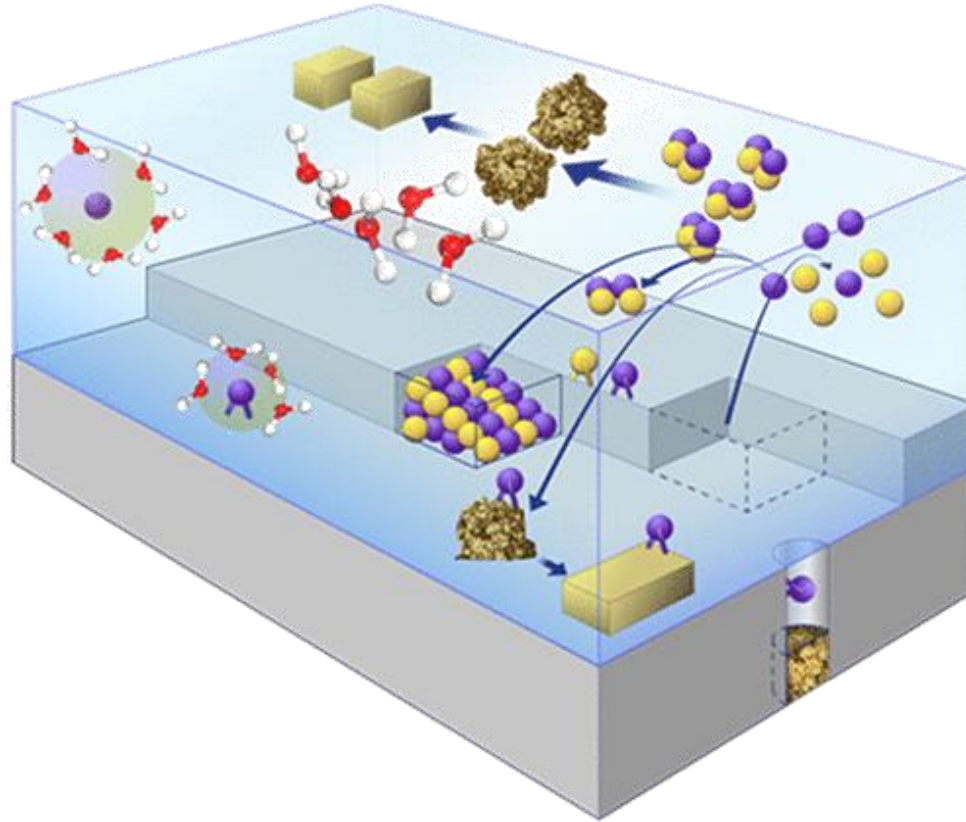
August 5, 2026



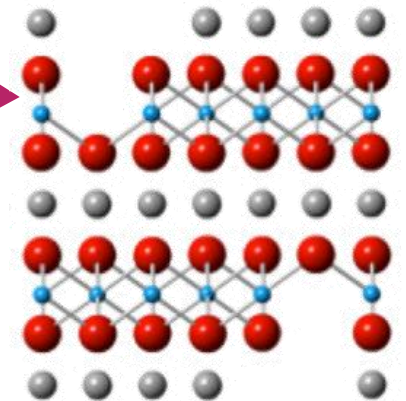
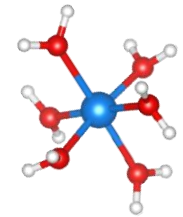
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Metal Release: A Key Environmental Challenge for Battery Materials

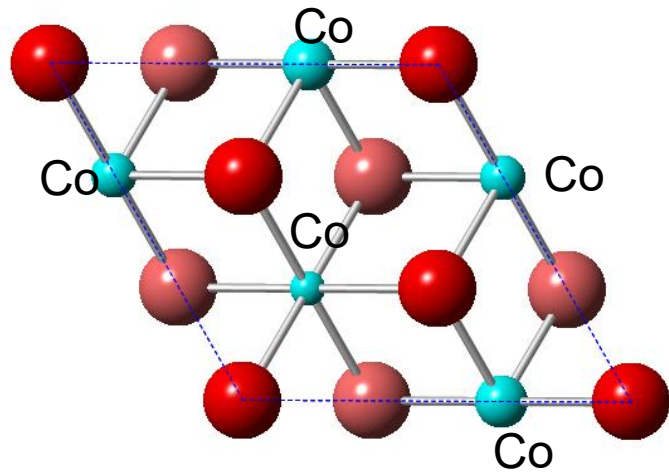


- The material starts to dissolve
- Bonds break
- Metals are released in water as cations

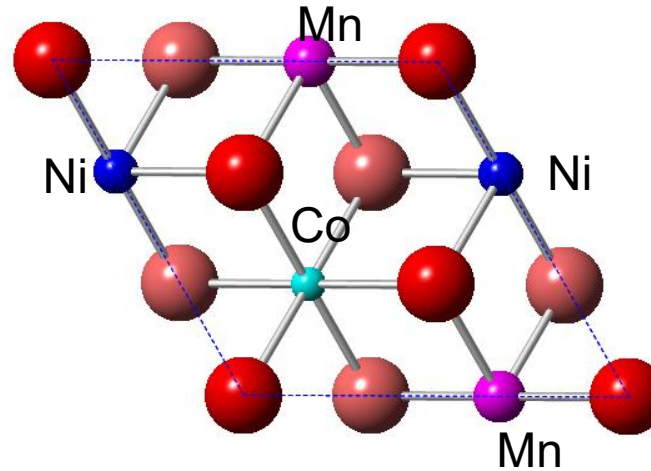


Designing Function Through Compositional Control: A Family of Oxides

LiCoO_2 is “LCO”



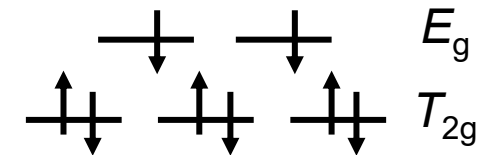
$\text{Li}(\text{Ni}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3})\text{O}_2$ is “NMC”



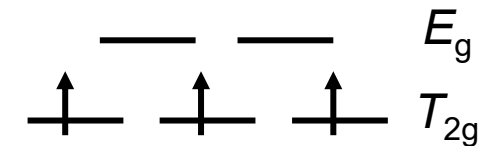
Changing the metals in the crystal changes the material's properties, including its tendency to release metals into the environment.

Our goal is to discover which material properties are most important so we can guide the design of better battery materials.

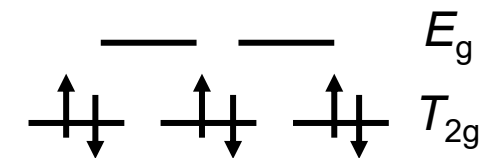
Ni^{2+} d^8 , spin = -1



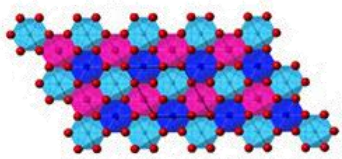
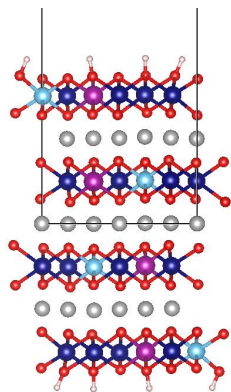
Mn^{4+} d^3 , spin = +3/2



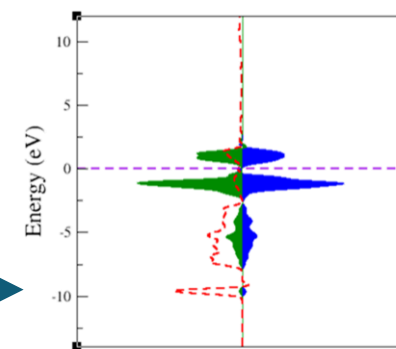
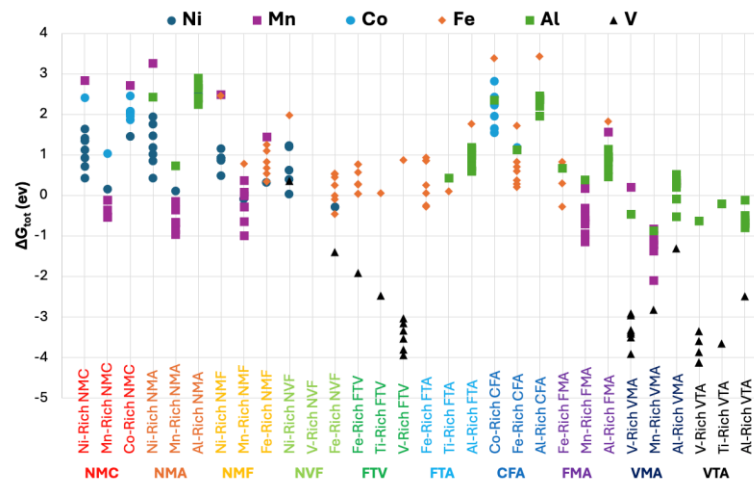
Co^{3+} d^6 , spin = 0



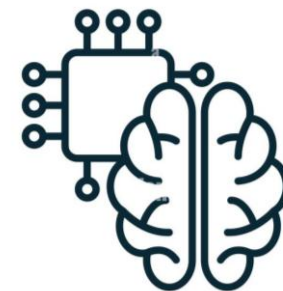
Beyond NMC...Same Structure, More Metal Variety



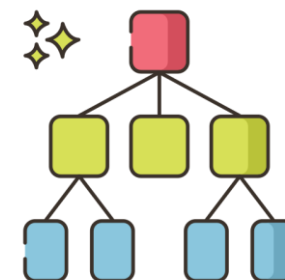
Ni, Mn, Co, Fe, Al, V, Ti



Extract
Descriptors

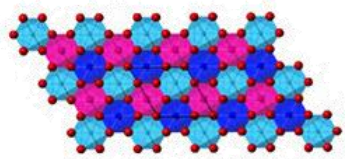
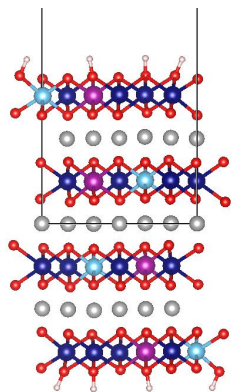


Machine
Learning
Regression

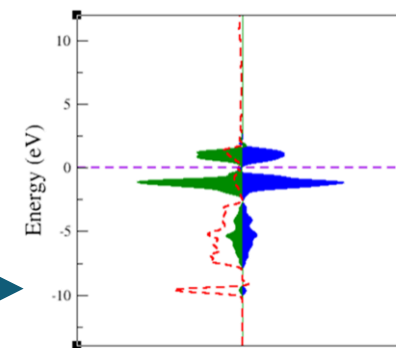
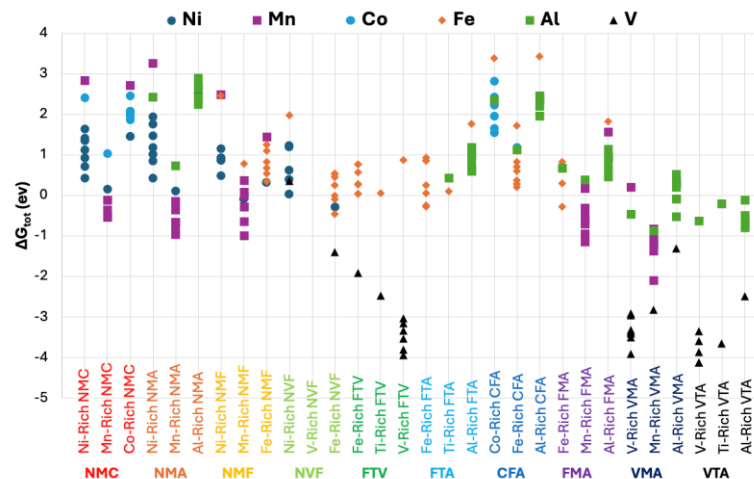


Today's activity uses a real research dataset developed at the Center for Functional Nanomaterials (CFN). Each DFT calculation shown here becomes one row in the machine learning dataset you'll analyze today.

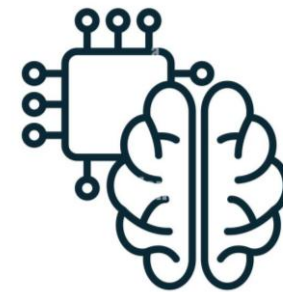
Beyond NMC...Same Structure, More Metal Variety



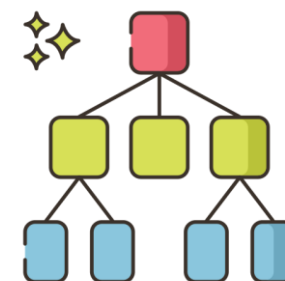
Ni, Mn, Co, Fe, Al, V, Ti



Extract
Descriptors



Machine
Learning
Regression



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C: PHYSICAL PROPERTIES OF MATERIALS AND INTERFACES | August 22, 2025

Fundamental Insights into Cathode Stability: Linking Compositional Tuning and Local Coordination in Complex Metal Oxides under Aqueous Transformations

Blake G. Hudson, Jennifer L. Bjorklund, Natalia C. Mena Santiago, Xiaohui Qu, Juan A. Santana, and Sara E. Mason*



**Natalia Mena
Santiago**
University of Puerto
Rico at Cayey
CFN Intern '24, '25

How Can AI Help Design More Sustainable Battery Materials?

Environmental Challenge

Lithium-ion batteries eventually reach **end of life**.

In aqueous environments, battery materials can release transition metal ions.

Understanding what controls **metal release** can help design safer, more environmentally sustainable materials.

Traditional Physics Approach

Density Functional Theory (DFT) predicts metal release energetics.

Accurate and physically grounded—but computationally expensive.

Scientific Question

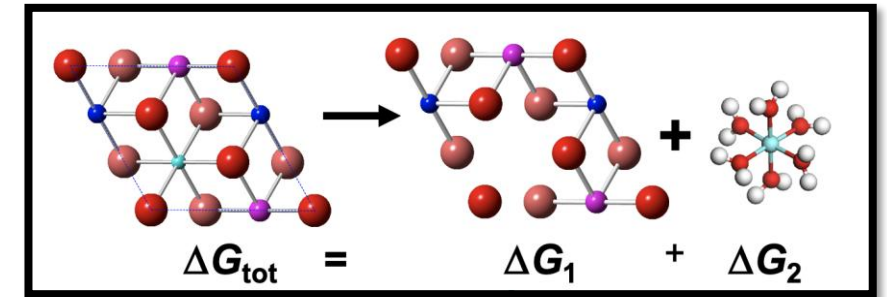
Can we identify a **small set of physically meaningful descriptors** that predict metal release energetics?

Machine Learning Approach

Train a **Random Forest machine learning model** using DFT-generated data to:

Predict metal release energetics (surrogate model)

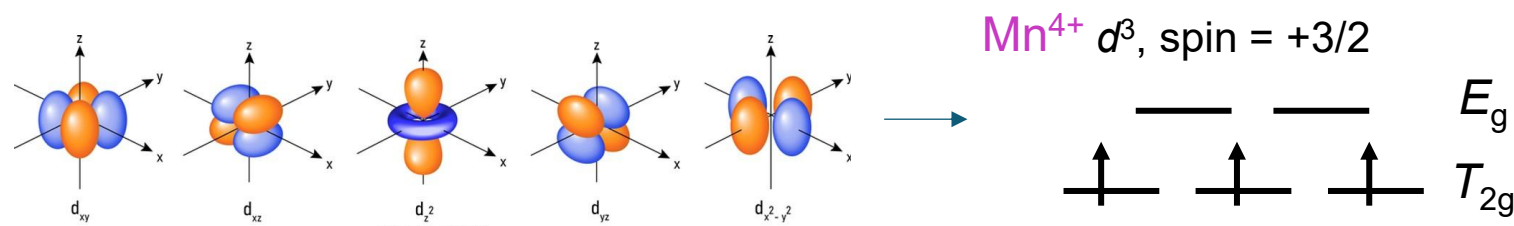
Identify the descriptors that matter most (hypothesis testing)



Today's Goal

Use AI not just to make predictions—but to gain scientific insight into what controls metal release from battery materials.

From Physics to Data



Every row in the dataset represents **one DFT calculation**.

Each row contains:

Battery Removed Target
Material Metal Property

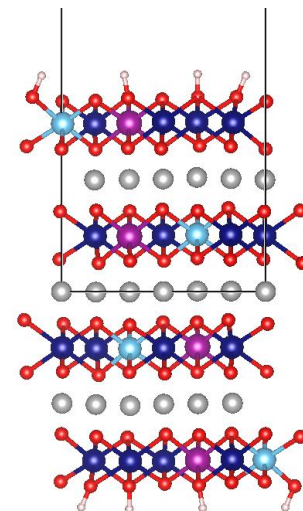
Descriptors

	Formulation	RemovedM	dGtotal	Descriptors						
				dG1	Z	Hybrid	OS	BV_vac	Charge_vac	Mag_vac
0	NiMnCo	Mn(1)	2.852267	7.793543	25	1.454092	3.5	3.921356	11.5152	2.2992
1	NiMnCo	Co(2)	2.408641	5.549917	27	0.824776	2.6	3.030442	13.9795	0.1396
2	NiMnCo	Ni(3)	1.122554	4.172830	28	0.549495	2.4	2.755209	15.0359	-0.8165
3	NiMnCo	Ni(4)	1.638729	4.689005	28	0.549495	2.4	2.755209	15.0359	-0.8165
4	NiMnCo	Ni(5)	0.436671	3.486947	28	0.397016	2.2	2.373854	15.0311	-1.2895

Challenge

Which descriptors contain enough information to predict metal release?

Designing a Scientific Hypothesis



Our Scientific Question

Can a small set of material descriptors predict metal release energy?

Our Initial Hypothesis

We believe the following descriptors contain enough information:

Scientific Idea	Descriptor Name	CSV Label
Ionic size of the metal	Ionic radius	Ionicradius
Bond strain/structural instability	Instability index	instabilityindex
Local oxygen bonding environment	Oxygen bond valence sum	BV_Oxygen

Before we run the model...

Prediction:

Will this hypothesis work?

Which descriptor do you think will be most important?

What Happens Next?

How Will We Judge Our Hypothesis?

The Random Forest model will answer two questions:

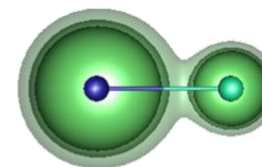
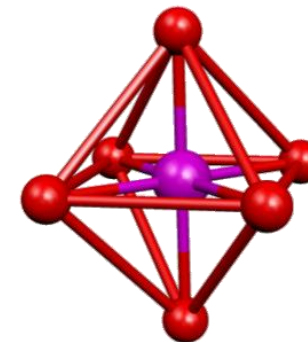
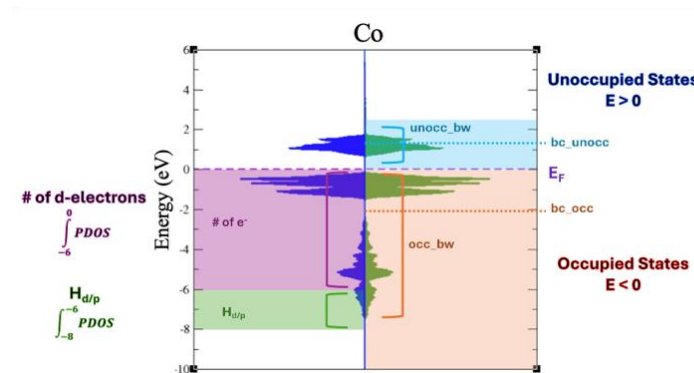
1. Can it make accurate predictions?

Compare predicted vs. actual metal release energies.

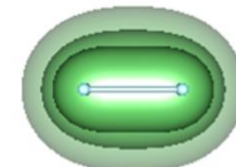
2. What did it learn?

Which descriptors were most important?

Does that agree with your prediction?



Ionic
Give and take the cloud



Covalent
Share the cloud

Your Turn: Test a Scientific Hypothesis

Return to **Section 4: Design Your Scientific Hypothesis**

Modify the descriptor list and rerun the cells below it.

Challenge 1

Single Descriptor

Test descriptors **one at a time** to discover which individual descriptor carries the most predictive information.

Question: Which single descriptor works best?

Challenge 2

Build a Better Hypothesis

Combine your strongest descriptors into a **3–5 descriptor model**. Does the performance improve?

Question: Can you improve upon the initial hypothesis?

Challenge 3

Your Scientific Hypothesis

Develop and test your own scientific explanation for metal release.

Question: What does your model teach you about the chemistry?

Record Your Results

Descriptor list

R² score

Most important descriptor

Did the data support your hypothesis?

Correlated Descriptors: More Isn't Always Better

Highly correlated descriptors often carry similar information.

Choosing descriptors that capture **different aspects of the chemistry** can improve both model performance and scientific insight.

Example	What the Heatmap Suggests	Why It Matters
Z ↔ Total_d_electrons, t2g, eg	Strong positive correlation	These descriptors all reflect where an element lies in the transition-metal series and its electronic structure. They may provide overlapping information.
Mag_env ↔ Mag_inst	Strong positive correlation	These descriptors describe related aspects of the local magnetic environment and are therefore not independent.
Hybrid ↔ Mag_inst	Weak correlation	These descriptors capture different physics (bonding vs. magnetism). Combining weakly correlated descriptors may provide complementary information.

Reference: Descriptor Definitions (1 of 2)

Descriptors are the scientific information provided to the machine learning model. Some are simple chemical properties (e.g., Z), while others are derived from DFT calculations and electronic structure analysis.

Descriptor	Meaning
Z	Atomic number of the metal being removed
OS	Oxidation state of the metal
Hybrid	Overlap of metal d and oxygen p orbitals
BV_vac	Bond valence sum of the MOH group before removal
Charge_vac	Charge sum of the MOH group before removal
Mag_vac	Sum of magnetic moments of the MOH group
Mag_env	Sum of magnetic moments of surrounding atoms (excluding the MOH group)
Mag_inst	Net magnetic moment of the site ($Mag_vac - Mag_env$)
BV_Oxygen	Bond valence sum of the oxygen atom being removed

Reference: Descriptor Definitions (2 of 2)

Descriptors are the scientific information provided to the machine learning model. Some are simple chemical properties (e.g., Z), while others are derived from DFT calculations and electronic structure analysis.

Descriptor	Meaning
Total_d_electrons	Total number of d electrons on the metal
d_up_tot	Number of spin-up d electrons
d_down_tot	Number of spin-down d electrons
d_diff	Difference between spin-up and spin-down d electrons
t2g	Number of d electrons occupying t_{2g} orbitals
eg	Number of d electrons occupying e_g orbitals
band_center_diff	Difference between the metal d -band center and oxygen p -band center
instability_index	Global Instability Index (GII): a bond-valence-derived measure of bond strain and structural instability
Ionicradius	Ionic radius of the metal (pm), assuming the specified oxidation state

**Thank you for joining us at
CFN today!**